

# Question ID 002dba45

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 002dba45

Line  $k$  is defined by  $y = -\frac{17}{3}x + 5$ . Line  $j$  is perpendicular to line  $k$  in the  $xy$ -plane. What is the slope of line  $j$ ?

Question ID 01682aa5

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 01682aa5

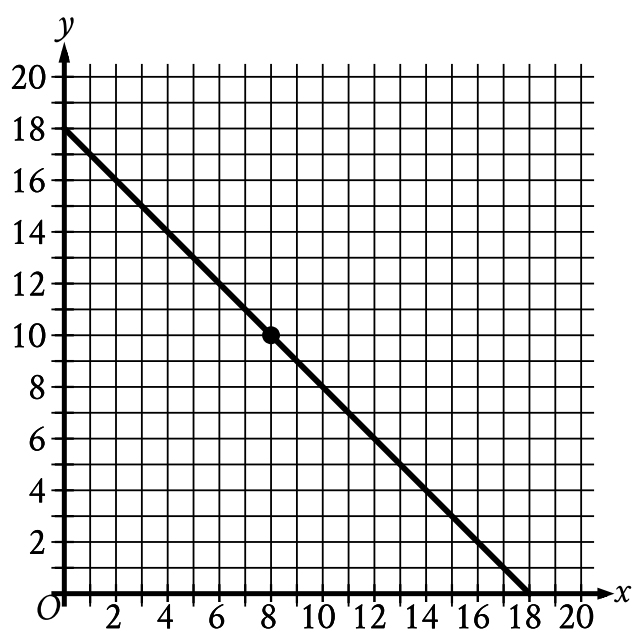
Line  $p$  is defined by  $2y + 18x = 9$ . Line  $r$  is perpendicular to line  $p$  in the  $xy$ -plane. What is the slope of line  $r$ ?

- A.  $-9$
- B.  $-\frac{1}{9}$
- C.  $\frac{1}{9}$
- D.  $9$

Question ID 9b0a4eae

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 9b0a4eae



The graph in the  $xy$ -plane models the possible combinations of length  $x$ , in meters (**m**), and width  $y$ , in meters, for a rectangle with a perimeter of **36 m**. Which statement is the best interpretation of the point **(8, 10)** in this context?

- A. The length is **10 m** less than the perimeter, and the width is **8 m** less than the perimeter.
- B. The length is **10 m**, and the width is **8 m**.
- C. The length is **8 m**, and the width is **10 m**.
- D. The length is **8 m** less than the perimeter, and the width is **10 m** less than the perimeter.

Question ID 0451d754

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 0451d754

Line  $k$  is defined by  $y = \frac{17}{7}x + 4$ . Line  $j$  is parallel to line  $k$  in the  $xy$ -plane. What is the slope of line  $j$ ?

- A.  $\frac{7}{17}$
- B.  $\frac{17}{7}$
- C. 4
- D. 17

# Question ID 1087f6c4

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 1087f6c4

$$24.5x + 24.75y = 641$$

Isabel ordered topsoil and crushed stone, which cost a total of ~~\$641~~, for her garden. The given equation represents the relationship between the number of cubic yards of topsoil,  $x$ , and the number of tons of crushed stone,  $y$ , Isabel ordered. How much more, in dollars, did a ton of crushed stone cost Isabel than a cubic yard of topsoil?

# Question ID 3c03cbd8

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 3c03cbd8

A certain township consists of a **5**-hectare industrial park and a **24**-hectare neighborhood. The total number of trees in the township is **4,529**. The equation  $5x + 24y = 4,529$  represents this situation. Which of the following is the best interpretation of  $x$  in this context?

- A. The average number of trees per hectare in the industrial park
- B. The average number of trees per hectare in the neighborhood
- C. The total number of trees in the industrial park
- D. The total number of trees in the neighborhood

# Question ID 7625073d

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 7625073d

The equation  $7g + 7b = 840$  represents the number of blue tiles,  $b$ , and the number of green tiles,  $g$ , an artist needs for an 840-square-inch tile project. The artist needs 71 blue tiles for the project. How many green tiles does he need?

# Question ID c5e38487

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: c5e38487

A chemist combines water and acetic acid to make a mixture with a volume of **56 milliliters (mL)**. The volume of acetic acid in the mixture is **10 mL**. What is the volume of water, in **mL**, in the mixture? (Assume that the volume of the mixture is the sum of the volumes of water and acetic acid before they were mixed.)

Question ID 9f3cb472

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 9f3cb472

Line  $t$  in the  $xy$ -plane has a slope of  $-\frac{1}{3}$  and passes through the point  $(9, 10)$ . Which equation defines line  $t$ ?

- A.  $y = 13x - \frac{1}{3}$
- B.  $y = 9x + 10$
- C.  $y = -\frac{x}{3} + 10$
- D.  $y = -\frac{x}{3} + 13$

# Question ID d7c8ba0b

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: d7c8ba0b

In the  $xy$ -plane, line  $t$  passes through the points  $(0, 9)$  and  $(1, 17)$ . Which equation defines line  $t$ ?

- A.  $y = \frac{1}{8}x + 9$
- B.  $y = x + \frac{1}{8}$
- C.  $y = x + 8$
- D.  $y = 8x + 9$

# Question ID 99ea3715

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 99ea3715

If the graph of  $27x + 33y = 297$  is shifted down 5 units in the  $xy$ -plane, what is the  $y$ -intercept of the resulting graph?

- A. (0, 4)
- B. (0, 6)
- C. (0, 14)
- D. (0, 28)

# Question ID 51568fb9

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 51568fb9

What is the slope of the graph of  $10x - 5y = -12$  in the  $xy$ -plane?

- A.  $-2$
- B.  $-\frac{5}{6}$
- C.  $\frac{5}{6}$
- D.  $2$

# Question ID 9c7741c6

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 9c7741c6

On a 210-mile trip, Cameron drove at an average speed of 60 miles per hour for the first  $x$  hours. He then completed the trip, driving at an average speed of 50 miles per hour for the remaining  $y$  hours. If  $x = 1$ , what is the value of  $y$  ?

# Question ID bbb0359a

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: bbb0359a

|     |    |    |    |
|-----|----|----|----|
| $x$ | 1  | 2  | 3  |
| $y$ | 11 | 16 | 21 |

The table shows three values of  $x$  and their corresponding values of  $y$ . Which equation represents the linear relationship between  $x$  and  $y$ ?

- A.  $y = 5x + 6$
- B.  $y = 5x + 11$
- C.  $y = 6x + 5$
- D.  $y = 6x + 11$

# Question ID 606cdce7

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 606cdce7

| $x$ | $y$ |
|-----|-----|
| −6  | 65  |
| −3  | 56  |
| 3   | 38  |
| 6   | 29  |

The table shows four values of  $x$  and their corresponding values of  $y$ . There is a linear relationship between  $x$  and  $y$ . Which of the following equations represents this relationship?

- A.  $9x + 3y = 141$
- B.  $9x + 3y = 3$
- C.  $3x + 9y = 141$
- D.  $3x + 9y = 3$

Question ID 483d208d

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 483d208d

At a state fair, attendees can win tokens that are worth a different number of points depending on the shape. One attendee won  $S$  square tokens and  $C$  circle tokens worth a total of  $1,120$  points. The equation  $80S + 90C = 1,120$  represents this situation. How many more points is a circle token worth than a square token?

- A. 950
- B. 90
- C. 80
- D. 10

# Question ID e0f59119

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

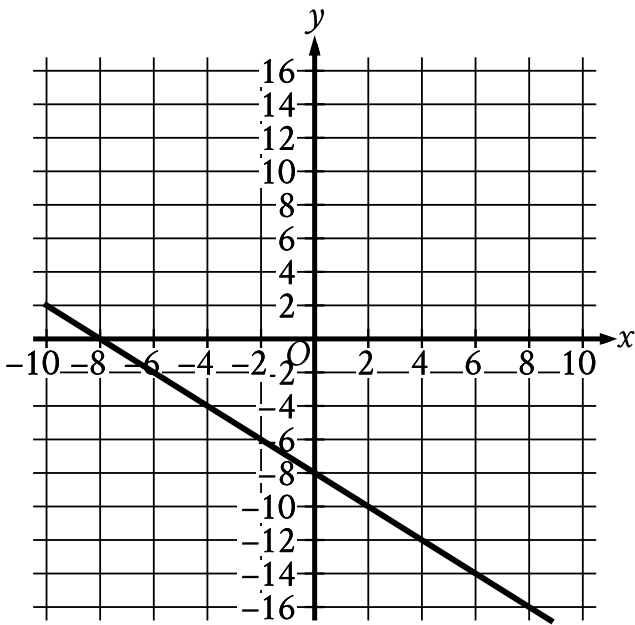
ID: e0f59119

What is the slope of the graph of  $y = \frac{1}{3}(29x + 10) + 5x$  in the  $xy$ -plane?

Question ID c307283c

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: c307283c



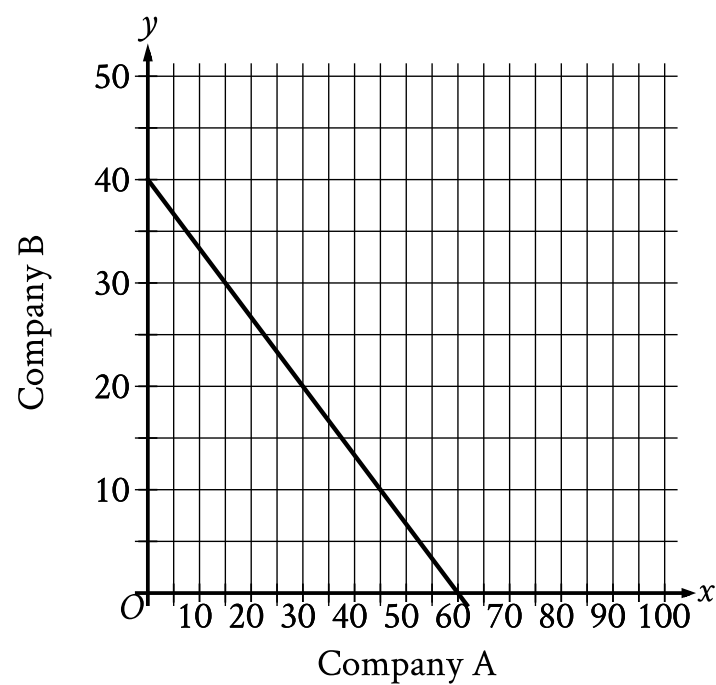
What is an equation of the graph shown?

- A.  $y = -2x - 8$
- B.  $y = x - 8$
- C.  $y = -x - 8$
- D.  $y = 2x - 8$

Question ID 2e0290c3

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 2e0290c3



The graph shows the relationship between the number of shares of stock from Company A,  $x$ , and the number of shares of stock from Company B,  $y$ , that Simone can purchase. Which equation could represent this relationship?

- A.  $y = 8x + 12$
- B.  $8x + 12y = 480$
- C.  $y = 12x + 8$
- D.  $12x + 8y = 480$

# Question ID df78b361

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: df78b361

Lily made **36** cups of jam. Lily then filled  **$x$**  small containers and  **$y$**  large containers with all the jam she made. The equation  **$4x + 6y = 36$**  represents this situation. Which is the best interpretation of  **$6y$**  in this context?

- A. The number of large containers Lily filled
- B. The number of small containers Lily filled
- C. The total number of cups of jam in the large containers
- D. The total number of cups of jam in the small containers

# Question ID 0d1b1e35

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 0d1b1e35

A batch of banana milkshakes consists of **4** cups of ice cream and **2** bananas and has **1,114 milligrams (mg)** of calcium. There is **276 mg** of calcium in **1** cup of the ice cream used to make this batch of milkshakes. How much calcium, **in mg**, is in **1** banana?

- A. 5
- B. 10
- C. 419
- D. 1,104

# Question ID d62ad380

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: d62ad380

An artist paints and sells square tiles. The selling price  $P$ , in dollars, of a painted tile is a linear function of the side length of the tile  $s$ , in inches, as shown in the table below.

| Side length, $s$ (inches) | Price, $P$ (dollars) |
|---------------------------|----------------------|
| 3                         | 8.00                 |
| 6                         | 18.00                |
| 9                         | 28.00                |

Which of the following could define the relationship between  $s$  and  $P$  ?

- A.  $P = 3s + 10$
- B.  $P = \frac{10}{3}s + 8$
- C.  $P = \frac{10}{3}s - 2$
- D.  $P = \frac{3}{10}s - \frac{1}{10}$

Question ID 92aa3a94

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 92aa3a94

Line  $k$  is defined by  $y = 7x + \frac{1}{8}$ . Line  $j$  is perpendicular to line  $k$  in the  $xy$ -plane. What is the slope of line  $j$ ?

- A.  $-8$
- B.  $-\frac{1}{7}$
- C.  $\frac{1}{8}$
- D.  $7$

# Question ID 431c3038

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 431c3038

In an article about exercise, it is estimated that a 160-pound adult uses 200 calories for every 30 minutes of hiking and 150 calories for every 30 minutes of bicycling. An adult who weighs 160 pounds has completed 1 hour of bicycling. Based on the article, how many hours should the adult hike to use a total of 1,900 calories from bicycling and hiking?

- A. 9.5
- B. 8.75
- C. 6
- D. 4

# Question ID 62ef6f73

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 62ef6f73

A total of **2** squares each have side length  $r$ . A total of **6** equilateral triangles each have side length  $t$ . None of these squares and triangles shares a side. The sum of the perimeters of all these squares and triangles is **210**. Which equation represents this situation?

- A.  $6r + 24t = 210$
- B.  $2r + 6t = 210$
- C.  $8r + 18t = 210$
- D.  $6r + 2t = 210$

# Question ID 265f2a53

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 265f2a53

When line  $n$  is graphed in the  $xy$ -plane, it has an  $x$ -intercept of  $(-4, 0)$  and a  $y$ -intercept of  $(0, \frac{86}{3})$ . What is the slope of line  $n$ ?

- A.  $\frac{3}{344}$
- B.  $\frac{6}{43}$
- C.  $\frac{43}{6}$
- D.  $\frac{344}{3}$

# Question ID f81a0503

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: f81a0503

In the  $xy$ -plane, line  $k$  passes through the points  $(0, -5)$  and  $(1, -1)$ . Which equation defines line  $k$ ?

- A.  $y = -x + \frac{1}{4}$
- B.  $y = \frac{1}{4}x - 5$
- C.  $y = -x + 4$
- D.  $y = 4x - 5$

# Question ID c39dbbdf

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: c39dbbdf

Line  $r$  is defined by the equation  $4x - 9y = 3$ . Line  $s$  is parallel to line  $r$  in the  $xy$ -plane. What is the slope of line  $s$ ?

- A.  $\frac{9}{4}$
- B.  $\frac{4}{9}$
- C.  $-4$
- D.  $-9$

# Question ID ca452900

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: ca452900

What is the slope of the graph of  $y = \frac{5x}{13} - 23$  in the  $xy$ -plane?

# Question ID 2e1a7f66

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 2e1a7f66

Figure A and figure B are both regular polygons. The sum of the perimeter of figure A and the perimeter of figure B is **63** inches. The equation  **$3x + 6y = 63$**  represents this situation, where  **$x$**  is the number of sides of figure A and  **$y$**  is the number of sides of figure B. Which statement is the best interpretation of **6** in this context?

- A. Each side of figure B has a length of **6** inches.
- B. The number of sides of figure B is **6**.
- C. Each side of figure A has a length of **6** inches.
- D. The number of sides of figure A is **6**.

# Question ID 637022d2

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 637022d2

$2.5b + 5r = 80$

The given equation describes the relationship between the number of birds,  $b$ , and the number of reptiles,  $r$ , that can be cared for at a pet care business on a given day. If the business cares for **16** reptiles on a given day, how many birds can it care for on this day?

- A. **0**
- B. **5**
- C. **40**
- D. **80**

# Question ID 6f6dfe3e

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 6f6dfe3e

| $x$  | $y$       |
|------|-----------|
| $-6$ | $n + 184$ |
| $-3$ | $n + 92$  |
| $0$  | $n$       |

The table shows three values of  $x$  and their corresponding values of  $y$ , where  $n$  is a constant, for the linear relationship between  $x$  and  $y$ . What is the slope of the line that represents this relationship in the  $xy$ -plane?

- A.  $-\frac{92}{3}$
- B.  $-\frac{3}{92}$
- C.  $\frac{n+92}{-3}$
- D.  $\frac{2n-92}{3}$

# Question ID 038d87d7

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 038d87d7

A neighborhood consists of a **2**-hectare park and a **35**-hectare residential area. The total number of trees in the neighborhood is **3,934**. The equation  **$2x + 35y = 3,934$**  represents this situation. Which of the following is the best interpretation of  $x$  in this context?

- A. The average number of trees per hectare in the park
- B. The average number of trees per hectare in the residential area
- C. The total number of trees in the park
- D. The total number of trees in the residential area

# Question ID 400798d6

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 400798d6

$$2x + y = 37$$

In triangle  $QRS$ , sides  $QR$  and  $RS$  each have a length of  $x$  centimeters and side  $SQ$  has a length of  $y$  centimeters. The given equation represents this situation. Which of the following is the best interpretation of  $37$  in this context?

- A. The difference, in centimeters, between the lengths of sides  $QR$  and  $SQ$
- B. The difference, in centimeters, between the lengths of sides  $QR$  and  $RS$
- C. The sum of the lengths, in centimeters, of the three sides of the triangle
- D. The length, in centimeters, of one of the two sides of equal length

# Question ID 9ed4c1a2

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 9ed4c1a2

What is the slope of the graph of  $y = \frac{1}{4}(27x + 15) + 7x$  in the  $xy$ -plane?

# Question ID fb43b85f

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: fb43b85f

A line passes through the points  $(4, 6)$  and  $(15, 24)$  in the  $xy$ -plane. What is the slope of the line?

# Question ID a4989860

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: a4989860

A line in the  $xy$ -plane has a slope of  $9$  and passes through the point  $(0, -5)$ . The equation  $y = px + r$  defines the line, where  $p$  and  $r$  are constants. What is the value of  $p$ ?

Question ID 63be6c9a

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 63be6c9a

In the  $xy$ -plane, line  $s$  passes through the point  $(0, 0)$  and is parallel to the line represented by the equation  $y = 18x + 2$ . If line  $s$  also passes through the point  $(4, d)$ , what is the value of  $d$ ?

- A. 2
- B. 18
- C. 72
- D. 74

# Question ID a04190b7

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

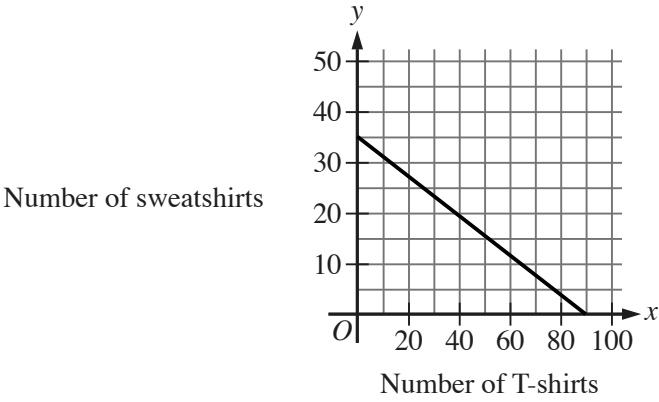
ID: a04190b7

A store sells two different-sized containers of blueberries. The store’s sales of these blueberries totaled **896.86** dollars last month. The equation  **$4.51x + 6.07y = 896.86$**  represents this situation, where  **$x$**  is the number of smaller containers sold and  **$y$**  is the number of larger containers sold. According to the equation, what is the price, in dollars, of each smaller container?

Question ID 00b9bd37

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 00b9bd37



The graph models the relationship between the number of T-shirts,  $x$ , and the number of sweatshirts,  $y$ , that Kira can purchase for a school fundraiser. Which equation could represent this relationship?

- A.  $y = 7x + 18$
- B.  $7x + 18y = 630$
- C.  $y = 18x + 7$
- D.  $18x + 7y = 630$

# Question ID 3008cfc3

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 3008cfc3

| $x$     | $y$   |
|---------|-------|
| $k$     | 13    |
| $k + 7$ | $-15$ |

The table gives the coordinates of two points on a line in the  $xy$ -plane. The  $y$ -intercept of the line is  $(k - 5, b)$ , where  $k$  and  $b$  are constants. What is the value of  $b$ ?

Question ID 3cdbf026

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 3cdbf026

The graph of the equation  $ax + ky = 6$  is a line in the  $xy$ -plane, where  $a$  and  $k$  are constants. If the line contains the points  $(-2, -6)$  and  $(0, -3)$ , what is the value of  $k$  ?

- A.  $-2$
- B.  $-1$
- C.  $2$
- D.  $3$

# Question ID 00723d16

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 00723d16

Line  $\ell$  is defined by  $3y + 12x = 5$ . Line  $n$  is perpendicular to line  $\ell$  in the  $xy$ -plane. What is the slope of line  $n$ ?

Question ID 9bbce683

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 9bbce683

| $x$ | $y$ |
|-----|-----|
| 18  | 130 |
| 23  | 160 |
| 26  | 178 |

For line  $h$ , the table shows three values of  $x$  and their corresponding values of  $y$ . Line  $k$  is the result of translating line  $h$  down 5 units in the  $xy$ -plane. What is the  $x$ -intercept of line  $k$ ?

- A.  $(-\frac{26}{3}, 0)$
- B.  $(-\frac{9}{2}, 0)$
- C.  $(-\frac{11}{3}, 0)$
- D.  $(-\frac{17}{6}, 0)$

# Question ID 686b7244

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 686b7244

A certain apprentice has enrolled in **85** hours of training courses. The equation  **$10x + 15y = 85$**  represents this situation, where  **$x$**  is the number of on-site training courses and  **$y$**  is the number of online training courses this apprentice has enrolled in. How many more hours does each online training course take than each on-site training course?

# Question ID db422e7f

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: db422e7f

Line  $p$  is defined by  $4y + 8x = 6$ . Line  $r$  is perpendicular to line  $p$  in the  $xy$ -plane. What is the slope of line  $r$ ?

# Question ID fdee0fbf

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: fdee0fbf

In the  $xy$ -plane, line  $k$  intersects the  $y$ -axis at the point  $(0, -6)$  and passes through the point  $(2, 2)$ . If the point  $(20, w)$  lies on line  $k$ , what is the value of  $w$  ?

# Question ID d0e614a6

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: d0e614a6

$$\frac{3}{5}x + \frac{3}{4}y = 7$$

Which table gives three values of  $x$  and their corresponding values of  $y$  for the given equation?

A.

| $x$ | $y$              |
|-----|------------------|
| 1   | $\frac{113}{20}$ |
| 2   | $\frac{101}{20}$ |
| 4   | $\frac{77}{20}$  |

B.

| $x$ | $y$            |
|-----|----------------|
| 1   | $\frac{47}{5}$ |
| 2   | $\frac{44}{5}$ |
| 4   | $\frac{38}{5}$ |

C.

| $x$ | $y$              |
|-----|------------------|
| 1   | $\frac{148}{15}$ |
| 2   | $\frac{136}{15}$ |
| 4   | $\frac{112}{15}$ |

D.

| $x$ | $y$              |
|-----|------------------|
| 1   | $\frac{128}{15}$ |
| 2   | $\frac{116}{15}$ |
| 4   | $\frac{92}{15}$  |

Question ID 9aaf7786

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 9aaf7786

In the  $xy$ -plane, line  $p$  has a slope of  $-\frac{5}{3}$  and an  $x$ -intercept of  $(-6, 0)$ . What is the  $y$ -coordinate of the  $y$ -intercept of line  $p$ ?

# Question ID b9835972

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

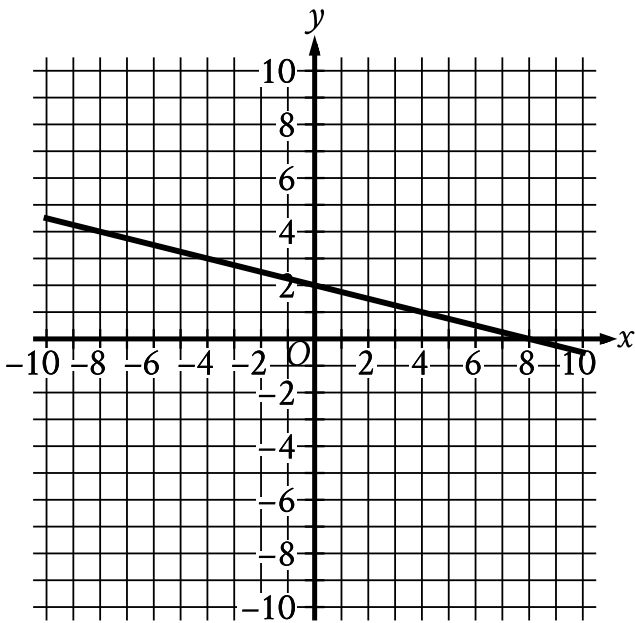
ID: b9835972

In the  $xy$ -plane, line  $\ell$  passes through the point  $(0, 0)$  and is parallel to the line represented by the equation  $y = 8x + 2$ . If line  $\ell$  also passes through the point  $(3, d)$ , what is the value of  $d$ ?

Question ID 05bb1af9

| Assessment | Test | Domain  | Skill                             | Difficulty                          |
|------------|------|---------|-----------------------------------|-------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div></div> <div></div> <div></div> |

ID: 05bb1af9



The graph of  $y = f(x) + 14$  is shown. Which equation defines function  $f$ ?

- A.  $f(x) = -\frac{1}{4}x - 12$
- B.  $f(x) = -\frac{1}{4}x + 16$
- C.  $f(x) = -\frac{1}{4}x + 2$
- D.  $f(x) = -\frac{1}{4}x - 14$

# Question ID cc7ffe02

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: cc7ffe02

Keenan made **32** cups of vegetable broth. Keenan then filled ***x*** small jars and ***y*** large jars with all the vegetable broth he made. The equation  **$3x + 5y = 32$**  represents this situation. Which is the best interpretation of  **$5y$**  in this context?

- A. The number of large jars Keenan filled
- B. The number of small jars Keenan filled
- C. The total number of cups of vegetable broth in the large jars
- D. The total number of cups of vegetable broth in the small jars

Question ID 0b46bad5

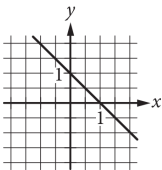
| Assessment | Test | Domain  | Skill                             | Difficulty                          |
|------------|------|---------|-----------------------------------|-------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div></div> <div></div> <div></div> |

ID: 0b46bad5

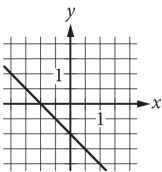
$ax + by = b$

In the equation above,  $a$  and  $b$  are constants and  $0 < a < b$ . Which of the following could represent the graph of the equation in the  $xy$ -plane?

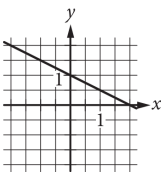
A.



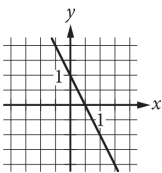
B.



C.



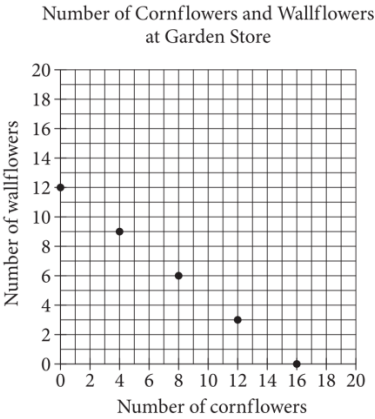
D.



# Question ID c362c210

| Assessment | Test | Domain  | Skill                             | Difficulty                          |
|------------|------|---------|-----------------------------------|-------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div></div> <div></div> <div></div> |

ID: c362c210



The points plotted in the coordinate plane above represent the possible numbers of wallflowers and cornflowers that someone can buy at the Garden Store in order to spend exactly \$24.00 total on the two types of flowers. The price of each wallflower is the same and the price of each cornflower is the same. What is the price, in dollars, of 1 cornflower?

Question ID 768f3b7c

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 768f3b7c

| $x$   | $y$ |
|-------|-----|
| $-2s$ | 24  |
| $-s$  | 21  |
| $s$   | 15  |

The table shows three values of  $x$  and their corresponding values of  $y$ , where  $s$  is a constant. There is a linear relationship between  $x$  and  $y$ . Which of the following equations represents this relationship?

- A.  $sx + 3y = 18s$
- B.  $3x + sy = 18s$
- C.  $3x + sy = 18$
- D.  $sx + 3y = 18$

Question ID 94b48cbf

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 94b48cbf

The graph of  $7x + 2y = -31$  in the  $xy$ -plane has an  $x$ -intercept at  $(a, 0)$  and a  $y$ -intercept at  $(0, b)$ , where  $a$  and  $b$  are constants. What is the value of  $\frac{b}{a}$ ?

- A.  $-\frac{7}{2}$
- B.  $-\frac{2}{7}$
- C.  $\frac{2}{7}$
- D.  $\frac{7}{2}$

# Question ID 98d3393a

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 98d3393a

Line  $\ell$  in the  $xy$ -plane is perpendicular to the line with equation  $x = 2$ . What is the slope of line  $\ell$ ?

- A. 0
- B.  $-\frac{1}{2}$
- C.  $-2$
- D. The slope of line  $\ell$  is undefined.

# Question ID cc3e9528

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: cc3e9528

The graph of  $9x - 10y = 19$  is translated down 4 units in the  $xy$ -plane. What is the  $x$ -coordinate of the  $x$ -intercept of the resulting graph?

# Question ID 0366d965

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 0366d965

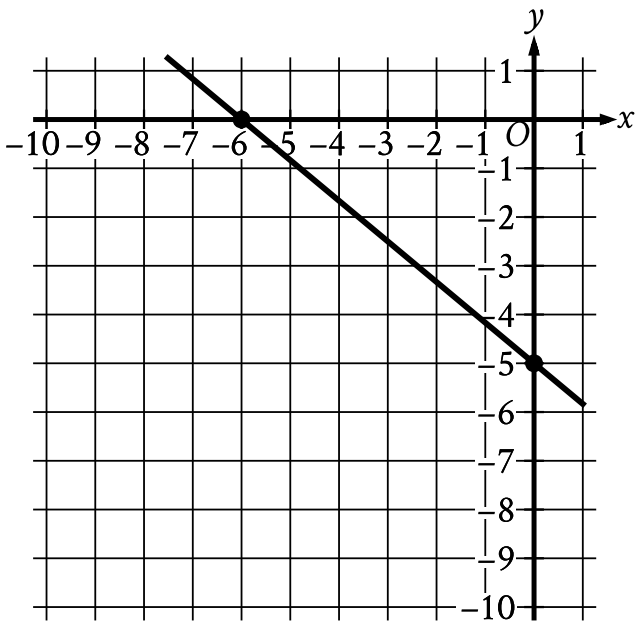
|     |     |
|-----|-----|
| $x$ | $y$ |
| 3   | 7   |
| $k$ | 11  |
| 12  | $n$ |

The table above shows the coordinates of three points on a line in the  $xy$ -plane, where  $k$  and  $n$  are constants. If the slope of the line is 2, what is the value of  $k+n$  ?

Question ID 6d8ad460

| Assessment | Test | Domain  | Skill                             | Difficulty                          |
|------------|------|---------|-----------------------------------|-------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div></div> <div></div> <div></div> |

ID: 6d8ad460



Line  $k$  is shown in the  $xy$ -plane. Line  $j$  (not shown) is perpendicular to line  $k$ . What is the slope of line  $j$ ?

Question ID 184ce5aa

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 184ce5aa

Line  $h$  is defined by  $\frac{1}{5}x + \frac{1}{7}y - 70 = 0$ . Line  $j$  is perpendicular to line  $h$  in the  $xy$ -plane. What is the slope of line  $j$ ?

- A.  $-\frac{7}{5}$
- B.  $-\frac{5}{7}$
- C.  $\frac{7}{5}$
- D.  $\frac{5}{7}$

# Question ID a35c7164

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: a35c7164

$$\begin{aligned}5x + 7y &= 1 \\ ax + by &= 1\end{aligned}$$

In the given pair of equations,  $a$  and  $b$  are constants. The graph of this pair of equations in the  $xy$ -plane is a pair of perpendicular lines. Which of the following pairs of equations also represents a pair of perpendicular lines?

- A.  $\begin{aligned}10x + 7y &= 1 \\ ax - 2by &= 1\end{aligned}$
- B.  $\begin{aligned}10x + 7y &= 1 \\ ax + 2by &= 1\end{aligned}$
- C.  $\begin{aligned}10x + 7y &= 1 \\ 2ax + by &= 1\end{aligned}$
- D.  $\begin{aligned}5x - 7y &= 1 \\ ax + by &= 1\end{aligned}$

# Question ID 2d54c272

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 2d54c272

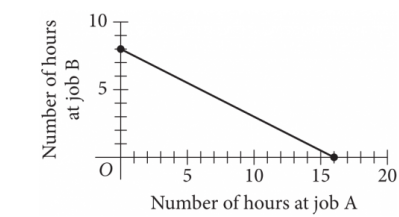
$$5G + 45R = 380$$

At a school fair, students can win colored tokens that are worth a different number of points depending on the color. One student won  $G$  green tokens and  $R$  red tokens worth a total of  $380$  points. The given equation represents this situation. How many more points is a red token worth than a green token?

Question ID c4ea43ef

| Assessment | Test | Domain  | Skill                             | Difficulty                          |
|------------|------|---------|-----------------------------------|-------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div></div> <div></div> <div></div> |

ID: c4ea43ef



To earn money for college, Avery works two part-time jobs: A and B. She earns \$10 per hour working at job A and \$20 per hour working at job B. In one week, Avery earned a total of  $s$  dollars for working at the two part-time jobs. The graph above represents all possible combinations of numbers of hours Avery could have worked at the two jobs to earn  $s$  dollars. What is the value of  $s$  ?

- A. 128
- B. 160
- C. 200
- D. 320

# Question ID cb58833c

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

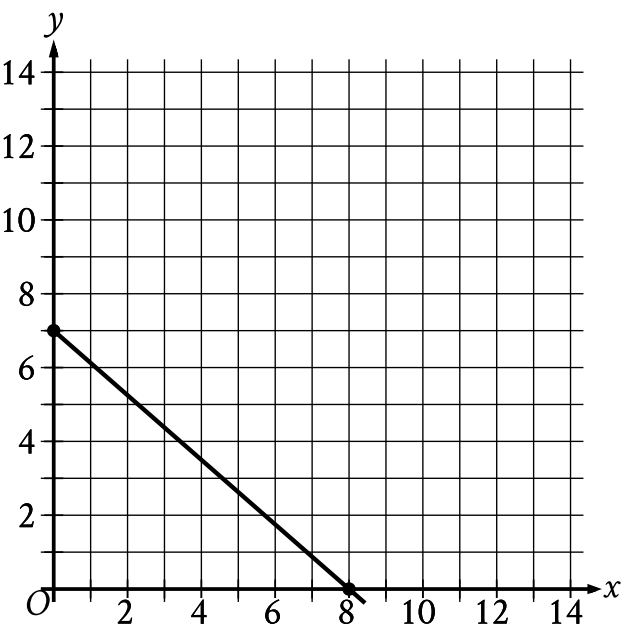
ID: cb58833c

The line with the equation  $\frac{4}{5}x + \frac{1}{3}y = 1$  is graphed in the  $xy$ -plane. What is the  $x$ -coordinate of the  $x$ -intercept of the line?

Question ID 9d0396d4

| Assessment | Test | Domain  | Skill                             | Difficulty                          |
|------------|------|---------|-----------------------------------|-------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div></div> <div></div> <div></div> |

ID: 9d0396d4



The point with coordinates  $(d, 4)$  lies on the line shown. What is the value of  $d$ ?

- A.  $\frac{7}{2}$
- B.  $\frac{26}{7}$
- C.  $\frac{24}{7}$
- D.  $\frac{27}{8}$

# Question ID a7a14e87

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: a7a14e87

In the  $xy$ -plane, line  $k$  is defined by  $x + y = 0$ . Line  $j$  is perpendicular to line  $k$ , and the  $y$ -intercept of line  $j$  is  $(0,3)$ . Which of the following is an equation of line  $j$  ?

- A.  $x + y = 3$
- B.  $x + y = -3$
- C.  $x - y = 3$
- D.  $x - y = -3$

# Question ID 9f70fd47

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 9f70fd47

What is the y-coordinate of the y-intercept of the graph of  $\frac{3x}{7} = -\frac{5y}{9} + 21$  in the xy-plane?

# Question ID a1fd2304

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

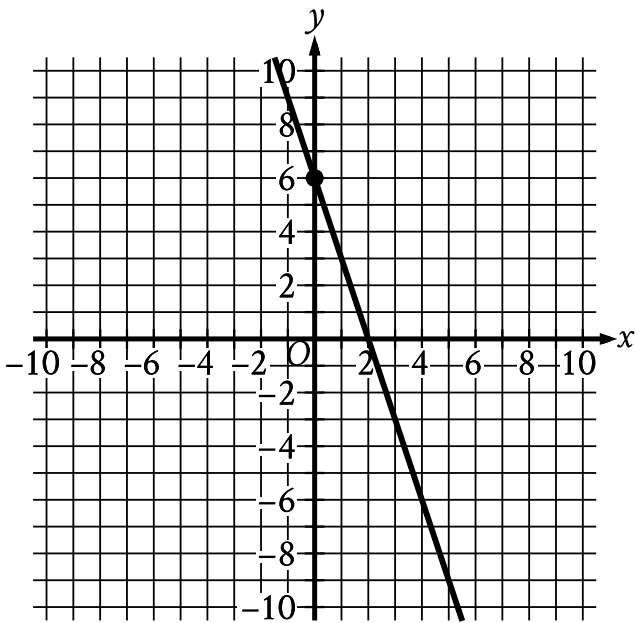
ID: a1fd2304

How many liters of a 25% saline solution must be added to 3 liters of a 10% saline solution to obtain a 15% saline solution?

Question ID 5b7599a6

| Assessment | Test | Domain  | Skill                             | Difficulty                          |
|------------|------|---------|-----------------------------------|-------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div></div> <div></div> <div></div> |

ID: 5b7599a6



The graph shows a linear relationship between  $x$  and  $y$ . Which equation represents this relationship, where  $R$  is a positive constant?

- A.  $Rx + 18y = 36$
- B.  $Rx - 18y = -36$
- C.  $18x + Ry = 36$
- D.  $18x - Ry = -36$

# Question ID 49800634

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 49800634

| $x$   | $y$   |
|-------|-------|
| $-18$ | $-48$ |
| $7$   | $52$  |

The table shows two values of  $x$  and their corresponding values of  $y$ . In the  $xy$ -plane, the graph of the linear equation representing this relationship passes through the point  $(\frac{1}{7}, a)$ . What is the value of  $a$ ?

- A.  $-\frac{4}{11}$
- B.  $-\frac{4}{77}$
- C.  $\frac{4}{7}$
- D.  $\frac{172}{7}$